

Tip. $\int_1^{\infty} \frac{\sinh x}{x} dx$ - ex. yce.

Опр. $]-\infty \leq a < b \leq +\infty$, $f \in \mathcal{L}_{\text{loc}}(a, b)$
 $\Rightarrow$ символ $\int_a^b f dx$ поз. и.и. $\text{or } f \text{ не } (a, b)$

Ans. $]-\infty \leq a < b \leq +\infty, f \in R_{loc}(a, b)$

$$\Rightarrow \underline{\text{Behaupte}} \quad \lim_{w_1 \rightarrow a+0} \int_{w_1}^c f \, dx + \lim_{w_2 \rightarrow b-0} \int_c^{w_2} f \, dx = J$$

если оба $\lim \exists b \in \mathbb{R}$ ($\neq \infty$ с разными знаками),
то значения И.Ч. $\neq$ не (a, b) ,

примем, если $T \in \mathbb{R}$, то и.ч. на свойстве,
иначе — р / сс — сс.

① $\left\{ \begin{array}{l} f, g \in \mathcal{R}_{loc} [a, b) \\ f \text{ dsc} - \underline{\text{osc. asc}} \end{array} \right. \Rightarrow$

① Leibniz $\int_a^b f(x) dx = \text{ex. ant.} \Rightarrow \int_a^b (f + g) dx = \text{ex. ant.}$

② $\int_a^b g(x) dx - \text{const.} \Rightarrow \int_a^b (f+g) dx - \text{const.}$

③ auch $\int_a^b g \, dx - p/cn \Rightarrow \int_a^b (f+g) \, dx - p/cn$

$$\Delta 1) \quad \int_a^b |f+g| dx \leq \int_a^b (|f| + |g|) dx =$$

$$= \underbrace{\int_a^b |f| dx}_{\text{ex.}} + \underbrace{\int_a^b |g| dx}_{\text{ex.}} \stackrel{\text{def}}{=} \int_a^b (|f| + |g|) dx - \text{ex.} \quad \bullet$$

$$2) \int_a^b (f+g) dx = \int_a^b f dx + \int_a^b g dx \quad (\text{сх. ас.} - \text{сх. ас.}) \Rightarrow$$

$$\Rightarrow \int_a^b [(f+g) - f] dx = \int_a^b g dx \quad \text{сх. ас.}$$

а это - правило с условием. $\Rightarrow \int_a^b (f+g) dx = \int_a^b f dx + \int_a^b g dx$

$$3) \int_a^b (f+g) dx = p / \text{сх.} \quad (\text{сх. ас.} \uparrow)$$

Опр. $\int f$ опр на $[a, b] \setminus \{c\}$, $c \in [a, b]$

$f \in R([a, \xi], [\eta, b])$, $\forall \xi, \eta$:
 $a \leq \xi < c < \eta \leq b$

$$\Rightarrow \int_a^b f dx \stackrel{\text{def}}{=} \lim_{\xi \rightarrow c-0} \int_a^{\xi} f dx + \lim_{\eta \rightarrow c+0} \int_{\eta}^b f dx \quad \text{н.н. ас. с известными значениями}$$

Если $\int \in \mathbb{R} \Rightarrow$ н.н. н.н. ас., иначе - п/сх.

Опр. $\int f$ опр. на $(a, b) \setminus \{c_k\}$, $k = 1, \dots, n$

$f \in R_{loc}([c_{i-1}, c_i])$, $-\infty \leq a < b \leq +\infty$

$$a = c_0 < c_1 < c_2 < \dots < c_n = b$$

$$\Rightarrow \int_a^b f dx \stackrel{\text{def}}{=} \sum_{i=1}^n \int_{c_{i-1}}^{c_i} f dx \quad \text{н.н. ас. или п/сх. на } [a, b]$$

Если $\int \in \mathbb{R}$, то н.н. н.н. ас., иначе - п/сх.

(в общем случае $\int \in \overline{\mathbb{R}}$)

Дифференцирование
 $\int_a^b f \cdot g dx$

- ① $f \in C^0[a, b]$
- ② $F(x) = \int_a^x f dx$ - опр. на $[a, b]$

③ $g \in C^1[a, b]$

④ g - монот. на $[a, b]$: $\begin{cases} g'(x) \leq 0 \\ g'(x) > 0 \end{cases} \quad x \in [a, b]$

⑤ $\lim_{x \rightarrow b-0} g(x) = 0$

$$\Rightarrow \int_a^b f \cdot g dx - \text{ср.}$$

Вспомогательный критерий Коши!

$$\forall \varepsilon > 0 \exists \Delta \in (a, b): \forall \sigma_1, \sigma_2: \begin{cases} \sigma_1 < \sigma_2 \\ \sigma_1 > \Delta \\ \sigma_2 > \Delta \end{cases}$$

$$\Rightarrow \left| \int_{\sigma_1}^{\sigma_2} f \cdot g dx \right| < \varepsilon ?$$

$$\int_{\sigma_1}^{\sigma_2} f \cdot g dx = \int_{\sigma_1}^{\sigma_2} g dF(x) = F(x) \cdot g(x) \Big|_{\sigma_1}^{\sigma_2} - \int_{\sigma_1}^{\sigma_2} F(x) \cdot g'(x) dx$$

M no var.

$$\begin{aligned} & \left| F(x) g(x) \Big|_{\sigma_1}^{\sigma_2} \right| = \left| F(x) \Big|_{\sigma_1}^{\sigma_2} \right| |g(\sigma_2) - g(\sigma_1)| \leq \\ & \leq M (|g(\sigma_2)| + |g(\sigma_1)|); \end{aligned}$$

$$\begin{aligned} \left| \int_{\sigma_1}^{\sigma_2} F(x) g'(x) dx \right| & \leq \int_{\sigma_1}^{\sigma_2} |F(x)| |g'(x)| dx \leq \\ & \leq M \left| \int_{\sigma_1}^{\sigma_2} \begin{bmatrix} -g(\sigma_2) + g(\sigma_2) \\ g(\sigma_2) - g(\sigma_1) \end{bmatrix} dx \right| = \end{aligned}$$

$$= M |g(\sigma_2) - g(\sigma_1)| \leq M (|g(\sigma_1)| + |g(\sigma_2)|)$$

$$\Rightarrow \left| \int_{\sigma_1}^{\sigma_2} f \cdot g dx \right| \leq 2M (|g(\sigma_1)| + |g(\sigma_2)|) < \varepsilon ?$$

$$\left\{ |A - B| = |A + (-B)| \leq |A| + |B|! \right\}$$

$$\text{Т.к. } \lim_{x \rightarrow b-0} g(x) = 0 \Rightarrow \forall \varepsilon > 0 \exists \delta = \delta(\varepsilon) > 0:$$

$$\forall x \in [\delta, b] \Rightarrow |g(x)| < \varepsilon \Rightarrow$$

$$\forall \sigma_1, \sigma_2 \in [\delta, b] \Rightarrow \begin{cases} |g(\sigma_1)| < \varepsilon \\ |g(\sigma_2)| < \varepsilon \end{cases}$$

$$\Rightarrow \left| \int_{D_1} f \cdot g \, dx \right| \leq 2M(\varepsilon + \varepsilon) = 4M\varepsilon = \tilde{\varepsilon}$$

$$\Rightarrow \int_a^b f \cdot g \, dx - \text{ср. по Кр. Коши. } \Delta$$

① Теорема Абеля.

$$\sup \int_a^b f \cdot g \, dx$$

① $f \in C^0[a, b]$

② $\int_a^b f \, dx - \text{ср.}$

③ $g \in C^1[a, b]$

④ g - монотонная на $[a, b]$

⑤ g - опр. на $[a, b]$

$$\Rightarrow \int_a^b f \cdot g \, dx - \text{ср.}$$

Δ Т.к. $\begin{cases} g - \text{опр. на } [a, b] \\ g - \text{монот.} \end{cases} \Rightarrow \exists \lim_{x \rightarrow b-0} g(x) \in \mathbb{R}$

$$\Rightarrow \sup g_1(x) = g(x) - g(b-0)$$

$$\sup \lim_{x \rightarrow b-0} g_1(x) = \lim_{x \rightarrow b-0} (g(x) - g(b-0)) = 0$$

Т.к. $\int_a^b f \, dx - \text{ср.} \Rightarrow F(\omega) = \int_a^\omega f \, dx - \text{опр.}$

Т.д. $\Rightarrow \int_a^b f \cdot g_1 \, dx - \text{ср.}, \sup f(x)g(x) =$

$$= f(a) \overline{g_1(a)} + f(a) \overline{g(b-0)}$$

$$\Rightarrow \int_a^b f \cdot g \, dx - \text{ср.} \quad \Delta \quad \text{ср. (как } \int \text{ Римана)}$$

Пр $\int_0^{+\infty} (e^x + x) \cos e^{2x} \, dx = J$

$\int e^{2x} = t \Rightarrow x = \frac{1}{2} \ln t, \quad dx = \frac{dt}{2t}$

$$\Rightarrow J = \frac{1}{2} \int_1^\infty \frac{\cos t}{\sqrt{t}} \, dt + \frac{1}{4} \int_1^\infty \frac{\ln t}{t} \cos t \, dt$$

$$J_1 = \int_1^{\infty} \frac{\cos t}{\sqrt{t}} dt \quad f(t) = \cos t \in C^0[1, \infty) \quad \int_1^{\infty} \cos t dt = \sin \infty - \sin 1 = \text{ср.} \quad \text{не } L[1, \infty)$$

$$g(t) = \frac{1}{\sqrt{t}}$$

$$g \in C^1[1, \infty) \quad (+)$$

$$\frac{1}{\sqrt{t}} - \text{монот. убав. не } L[1, \infty) \quad (+)$$

$$\lim_{t \rightarrow \infty} \frac{1}{\sqrt{t}} = 0 \quad (+)$$

$$\Rightarrow \text{но ср. Дир } J_1 - \text{ср.}$$

$$J_2 = \int_1^{\infty} \frac{\ln t}{t} \cos t dt \quad (+)$$

$$f(t) = \cos t \quad \text{ср-т } \text{убав. Дир.}$$

$$g(t) = \frac{\ln t}{t} \quad g'(t) = \frac{1 - \ln t}{t^2} \in C^0[1, \infty) \quad (+)$$

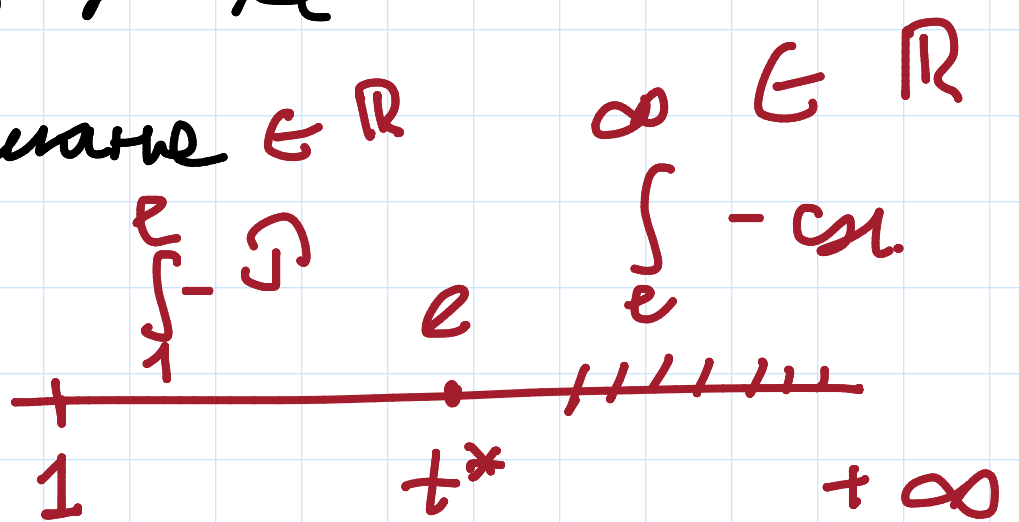
$$\exists t^* \in (1, \infty) : \forall t \gg t^* \quad g'(t) < 0 \quad (+)$$

$$\lim_{t \rightarrow \infty} \frac{\ln t}{t} = 0 \quad (+)$$

$$\Rightarrow J_2 = \int_1^{t^*} + \int_{t^*}^{\infty} - \text{ср.}$$

ср, г.к. ср. по Д-м

$$\text{Вывод: } J - \text{ср.}$$



Главное значение н.ч.

$$J \int_{-\infty}^{+\infty} x dx$$

$$J \lim_{\substack{a \rightarrow -\infty \\ b \rightarrow +\infty}} \int_a^b x dx = \frac{1}{2} \lim_{\substack{a \rightarrow -\infty \\ b \rightarrow +\infty}} (b^2 - a^2) = \{\infty - \infty\}$$

$$\text{нам} \int_{-\infty}^{+\infty} x dx = \int_{-\infty}^c x dx + \int_c^{+\infty} x dx = p/cx.$$

\quad \quad \quad \text{pacx.} \quad \quad \quad c \quad \quad \quad \text{pacx.}

$$\boxed{a = -b} \Rightarrow \int_{-\infty}^{+\infty} x dx = \lim_{a \rightarrow +\infty} \int_{-a}^a x dx = 0 \in \mathbb{R}$$

$a = (\frac{a^2}{2} - \frac{b^2}{2})$

$$0 = \lim_{a \rightarrow +\infty} \int_{-a}^a x dx = \text{v. p.} \int_{-\infty}^{+\infty} x dx$$

Valueur principal

Def. $\exists b \in \mathbb{R} \lim_{a \rightarrow +\infty} \int_{-a}^a f(x) dx,$

то он наз. главным значением и.и. $\int_{-\infty}^{+\infty} f dx.$
и об-ще. v. p. $\int_{-\infty}^{+\infty} f(x) dx.$

Зам. (1) $\exists \lim_{a \rightarrow +\infty} \int_{-\infty}^a f dx - \text{ср.} \Rightarrow \int_{-\infty}^{+\infty} f dx = \text{v. p.} \int_{-\infty}^{+\infty} f dx$

(2) $\exists \lim_{a \rightarrow +\infty} \int_{-\infty}^a f dx - p/cx$, то v. p. $\int_{-\infty}^{+\infty} f dx$
номер $\exists -T6.$

$\int_a^b f dx, f \in (R[a, b] | c), c \in [a, b]$
 $\xrightarrow{\text{def}} \int_a^b f dx = \lim_{\substack{\varepsilon_1 \rightarrow 0 \\ \varepsilon_2 \rightarrow 0}} \left[\int_a^{c-\varepsilon_1} f dx + \int_{c+\varepsilon_2}^b f dx \right]$

$$\boxed{\varepsilon_1 = \varepsilon_2}$$

Def $\exists b \in \mathbb{R} \lim_{\varepsilon \rightarrow 0} \left[\int_a^{c-\varepsilon} f dx + \int_{c+\varepsilon}^b f dx \right],$ то

этот $\lim$ наз. v. p. $\int_a^b f dx$, об-ще

Ex. $\int_{-1}^2 \frac{dx}{x}$, $x=0$ - a. r. - v. p. II type

$$\int_{-1}^2 \frac{dx}{x} = \int_{-1}^0 \frac{dx}{x} + \int_0^2 \frac{dx}{x} = \lim_{\varepsilon_1 \rightarrow 0} \int_{-1}^{-\varepsilon_1} \frac{dx}{x} +$$

$$+ \lim_{\varepsilon_2 \rightarrow 0} \int_{\varepsilon_2}^2 \frac{dx}{x} = \lim_{\varepsilon_1 \rightarrow 0} (\ln |-\varepsilon_1|) + \lim_{\varepsilon_2 \rightarrow 0} (\ln 2 - \ln \varepsilon_2)$$

$$= -\infty + \infty \Rightarrow \int_{-1}^2 \frac{dx}{x} - \text{p.l.c.}$$

$$\exists \varepsilon_1 = \varepsilon_2 = \varepsilon$$

$$\lim_{\varepsilon \rightarrow 0} \left(\int_{-1}^{-\varepsilon} \frac{dx}{x} + \int_{\varepsilon}^2 \frac{dx}{x} \right) = \lim_{\varepsilon \rightarrow 0} (\ln \varepsilon + \ln 2 - \ln \varepsilon)$$

$$= \ln 2 \Rightarrow \int_{-1}^2 \frac{dx}{x} - \text{p.l.c., too}$$

$$\exists \text{ v.p. } \int_{-1}^2 \frac{dx}{x} = \ln 2$$